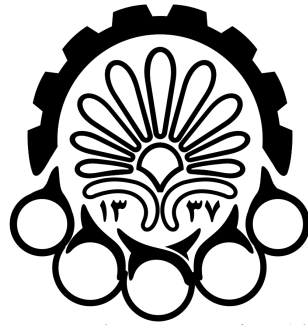




دانشگاه صنعتی امیرکبیر
(پلی تکنیک تهران)

دانشکده مهندسی کامپیوتر

درس مبانی رایانش ابری

فاز اول پروژه

نگارنده:

علی ناصری نیا

۴۰۱۳۱۰۵۳

محمد امین فرح بخش

۴۰۱۳۱۰۲۹

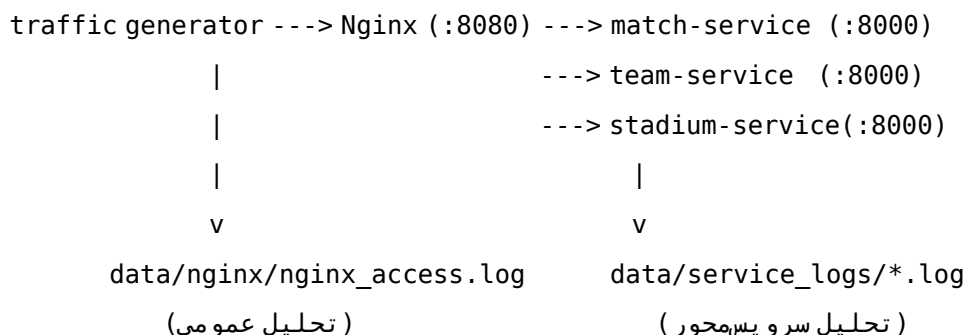
استاد راهنما:

دکتر جوادی

تیر ۱۴۰۵

معماری

اجرای پروژه بر اساس معماری زیر طراحی شده است:



نکات کلیدی معماری:

- پورت سرویس‌های backend روی هاست publish نشده است، بنابراین تولیدکننده‌ی ترافیک به‌هیچ‌وجه نمی‌تواند Nginx را دور بزند.
- Nginx هدرهای x-request-id، x-client-country و x-scenario را به سرویس مقصد منتقل می‌کند و همان‌ها در لاگ ساختاریافته‌ی سرویس ثبت می‌شوند.
- Nginx در \$request_time بر حسب ثانیه است و در Job 1 به request_time_ms تبدیل می‌شود.

اجرای موفق سرویس‌ها

سرویس match-service

```
~/Desktop/cloud/CC_Project_Phase1_40131053_40131029> http get "http://localhost:8080/api/matches?date=2026-06-25"
```

date	2026-06-25						
count	3						
matches	#	team_a	team_b	stadium	city	country	stage
	0	Netherlands	Morocco	BC Place Vancouver	Vancouver	Canada	Group Stage
	1	Belgium	Australia	Kansas City Stadium	Kansas City	USA	Group Stage
	2	Croatia	Italy	Miami Stadium	Miami	USA	Group Stage

شکل ۱: رفتار درست

```
~/Desktop/cloud/CC_Project_Phase1_40131053_40131029> http get "http://localhost:8080/api/matches?date=2026-06-"
Error: nu::shell::http_error

* HTTP Error 400 (Bad Request): http://localhost:8080/api/matches?date=2026-06-
[repl_entry #252:1:10]
1 http get "http://localhost:8080/api/matches?date=2026-06-"
.
.
.
{"error":"invalid date format, expected YYYY-MM-DD","received":"2026-06-"}

```

شکل ۲: تاریخ نامعتبر

سرویس team-service

```
~/Desktop/cloud/CC_Project_Phase1_40131053_40131029> http get "http://localhost:8080/api/teams?name=Argentina"
```

name	Argentina
country_code	ARG
confederation	CONMEBOL

شکل ۳: رفتار درست

```
~/Desktop/cloud/CC_Project_Phase1_40131053_40131029> http get "http://localhost:8080/api/teams?name=Argentin"
Error: nu::shell::http_error

  × HTTP Error 404 (Not Found): http://localhost:8080/api/teams?name=Argentin
  [repl_entry #268:1:10]
1 http get "http://localhost:8080/api/teams?name=Argentin"
  { "error": "team not found", "name": "Argentin" }
```

شکل ۴: کشور نامعتبر

سرویس stadium-service

```
~/Desktop/cloud/CC_Project_Phase1_40131053_40131029> let params = { name: "New York New Jersey Stadium" }
~/Desktop/cloud/CC_Project_Phase1_40131053_40131029> http get $"http://localhost:8080/api/stadiums?($params | url build-query)"
```

name	New York New Jersey Stadium
common_name	MetLife Stadium
city	New York New Jersey
country	USA
capacity	82500

شکل ۵: رفتار درست

```
~/Desktop/cloud/CC_Project_Phase1_40131053_40131029> let params = { name: "ghale hasan khan" }
~/Desktop/cloud/CC_Project_Phase1_40131053_40131029> http get $"http://localhost:8080/api/stadiums?($params | url build-query)"
Error: nu::shell::http_error

  × HTTP Error 404 (Not Found): http://localhost:8080/api/stadiums?name=ghale+hasan+khan
  [repl_entry #300:1:10]
1 http get $"http://localhost:8080/api/stadiums?($params | url build-query)"
  { "error": "stadium not found", "name": "ghale hasan khan" }
```

شکل ۶: ورزشگاه نامعتبر

اجرای موفق کانتینرها

خروجی دستور docker compose ps نشان‌دهنده‌ی اجرای صحیح تمام کانتینرها است:

#	NAME	IMAGE	COMMAND	SERVICE	CREATED	STATUS	PORTS
0	match-service	cc_project_phase1_40131053_40131029-match-service	"sh -c 'uvicorn main..."	match-service	2 hours ago	Up 2 hours	8000/tcp
1	nginx	nginx:1.25-alpine	"/docker-entrypoint..."	nginx	2 hours ago	Up 2 hours	0.0.0.0:8080->80/tcp, [::]:8080->80/tcp
2	stadium-service	cc_project_phase1_40131053_40131029-stadium-service	"sh -c 'uvicorn main..."	stadium-service	2 hours ago	Up 2 hours	8000/tcp
3	team-service	cc_project_phase1_40131053_40131029-team-service	"sh -c 'uvicorn main..."	team-service	2 hours ago	Up 2 hours	8000/tcp

شکل ۷: وضعیت کانتینرها

همچنین آزمون صحت تنظیمات Nginx با موفقیت انجام شد:

docker compose exec nginx nginx -t

```
~/Desktop/cloud/CC_Project_Phase1_40131053_40131029> docker compose exec nginx nginx -t
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful
```

شکل ۸: آزمون تنظیمات Nginx

تست درخواست از مسیر Nginx

جهت تست اولیه‌ی سیستم، یک درخواست نمونه به Nginx ارسال شد:

```
curl -H "X-Request-ID: manual-001" \
  -H "X-Client-Country: Iran" \
  -H "X-Scenario: normal" \
  "http://localhost:8080/api/teams?name=Argentina"
```

name	Argentina
country_code	ARG
confederation	CONMEBOL

شکل ۹: نتیجه‌ی درخواست تستی از مسیر nginx

اجرای traffic generator

برای تولید بار سنگین بر روی سیستم، از اسکریپت تولیدکننده‌ی ترافیک استفاده شد:

```
python3 traffic-generator/generate.py --requests 10000 --nginx-url http://localhost:8080
```

```
~/Desktop/cloud/CC_Project_Phase1_40131053_40131029> python3 traffic-generator/generate.py --requests 10000 --nginx-url http://localhost:8080
Sending 10000 requests to http://localhost:8080 with 32 workers ...
 5000 / 10000 done (16s)
10000 / 10000 done (30s)

Finished 10000 requests in 30.2s (331 req/s)
 2xx/3xx : 8741
 4xx    : 888
 5xx    : 371
 failed : 0

Nginx gateway log : data/nginx/nginx_access.log
Service logs      : data/service_logs/*.log
```

شکل ۱۰: اجرای تولیدکننده‌ی ترافیک با ۱۰۰۰۰۰ درخواست

بخشی از لاگ نهایی Nginx

پس از تولید ترافیک، لاگ Nginx با استفاده از دستور زیر مشاهده شد:

```
tail -n 3 data/nginx/nginx_access.log
```

```
~/Desktop/cloud/CC_Project_Phase1_40131053_40131029> tail -n 3 data/nginx/nginx_access.log | lines | each { from json }
```

#	timestamp	request_id	client_ip	client_country	scenario	method	path	service	status_code	request_time_sec	user_agent
0	2026-07-09T21:34:50+00:00	req_009908	172.18.0.1	Japan	slow	GET	/api/stadiums?name=Monterrey+Stadium	stadium-service	200	1.363	traffic-generator
1	2026-07-09T21:34:51+00:00	req_009977	172.18.0.1	Brazil	slow	GET	/api/stadiums?name=Atlanta+Stadium	stadium-service	200	1.267	traffic-generator
2	2026-07-09T21:34:51+00:00	req_009984	172.18.0.1	USA	slow	GET	/api/stadiums?name=BC+Place+Vancouver	stadium-service	200	1.459	traffic-generator

شکل ۱۱: سه خط آخر لاگ Nginx

بخشی از لاگ نهایی سرویس‌ها

به‌طور مشابه، لاگ سرویس‌ها نیز مشاهده شد:

```
tail -n 2 data/service_logs/team_service.log
```

```
tail -n 2 data/service_logs/match_service.log
```

```
tail -n 2 data/service_logs/stadium_service.log
```

```
~/Desktop/cloud/CC_Project_Phase1_40131053_40131029> tail -n 2 data/service_logs/team_service.log | lines | each { from json }
```

#	timestamp	request_id	client_country	scenario	service	endpoint	entity_type	entity_value	status_code	processing_time_ms	event_type
0	2026-07-09T21:34:50.010Z	req_009945	Iran	slow	team-service	/api/teams	team	Serbia	200	339	team_lookup
1	2026-07-09T21:34:50.492Z	req_009854	Canada	slow	team-service	/api/teams	team	Saudi Arabia	200	1058	team_lookup

شکل ۱۲: دو خط آخر لاگ team_service

```
~/Desktop/cloud/CC_Project_Phase1_40131053_40131029> tail -n 2 data/service_logs/match_service.log | lines | each { from json }
```

#	timestamp	request_id	client_country	scenario	service	endpoint	entity_type	entity_value	status_code	processing_time_ms	event_type
0	2026-07-09T21:34:50.391Z	req_009714	USA	slow	match-service	/api/matches	match_day	2026-06-13	200	1297	match_lookup
1	2026-07-09T21:34:50.574Z	req_009876	Brazil	slow	match-service	/api/matches	match_day	2026-06-22	200	1092	match_lookup

شکل ۱۳: دو خط آخر لاگ match-service

```
~/Desktop/cloud/CC_Project_Phase1_40131053_40131029> tail -n 2 data/service_logs/stadium_service.log | lines | each { from json }
```

#	timestamp	request_id	client_country	scenario	service	endpoint	entity_type	entity_value	status_code	processing_time_ms	event_type
0	2026-07-09T21:34:51.022Z	req_009977	Brazil	slow	stadium-service	/api/stadiums	stadium	Atlanta Stadium	200	1264	stadium_lookup
1	2026-07-09T21:34:51.025Z	req_009904	USA	slow	stadium-service	/api/stadiums	stadium	BC Place Vancouver	200	1457	stadium_lookup

شکل ۱۴: دو خط آخر لاگ stadium_service

اجرای Hadoop Streaming

برای پردازش داده‌ها، Hadoop Streaming به‌کار گرفته شد:

```
docker compose -f hadoop/docker-compose.yml up -d
```

```
docker exec -it namenode bash /project/scripts/run_mapreduce.sh
```

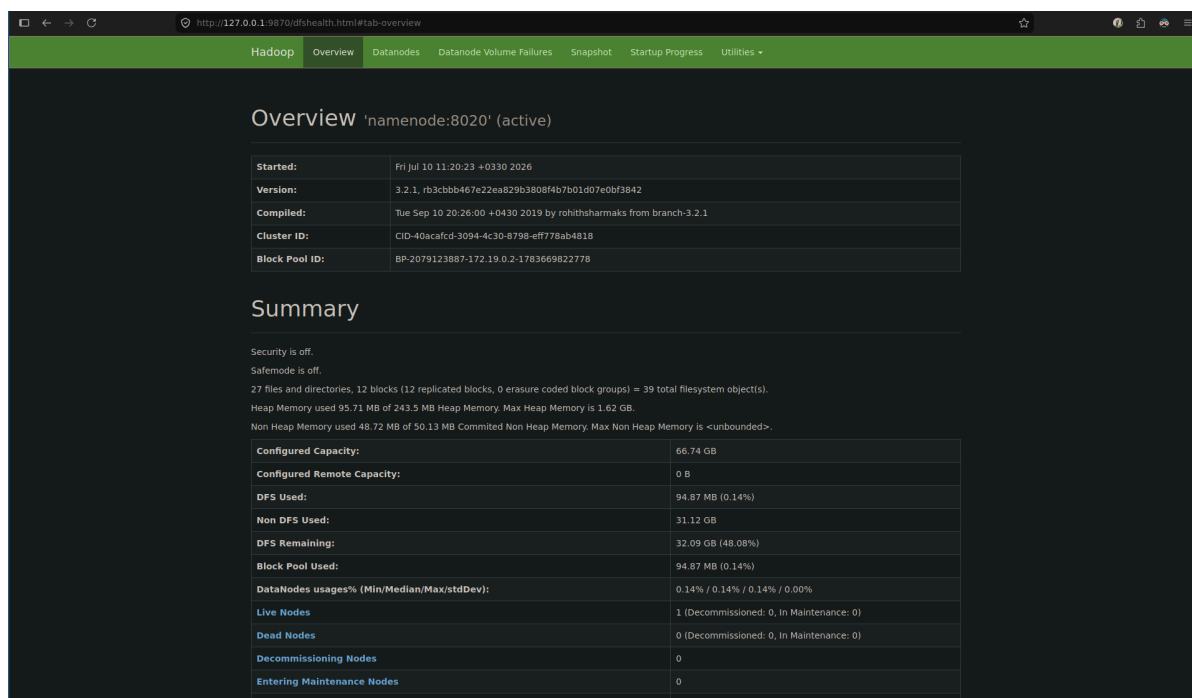
```
~/Desktop/cloud/CC_Project_Phase1_40131053_40131029> docker compose -f hadoop/docker-compose.yml up -d
[+] up 21/21
✓ Image registry.docker.ir/bde2020/hadoop-datanode:2.0.0-hadoop3.2.1-java8 Pulled
✓ Image registry.docker.ir/bde2020/hadoop-namenode:2.0.0-hadoop3.2.1-java8 Pulled
✓ Network hadoop_default Created
✓ Container namenode Started
✓ Container datanode Started
```

شکل ۱۵: ساخته شدن Hadoop

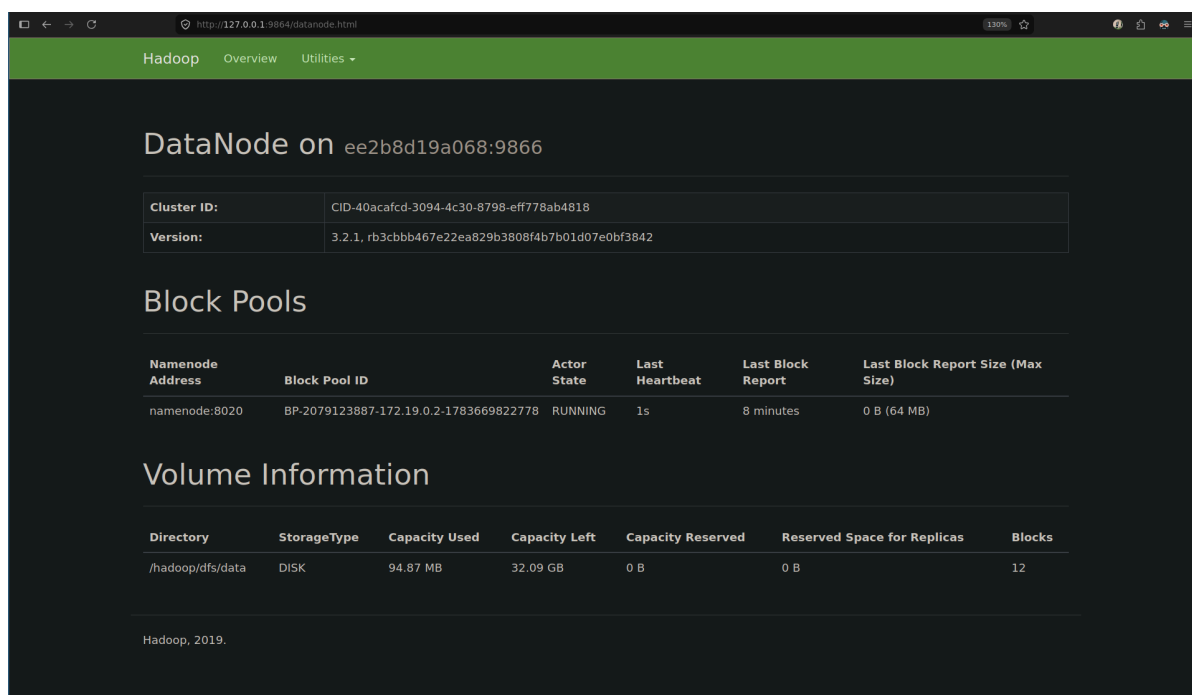
```
~/Desktop/cloud/CC_Project_Phase1_40131053_40131029> docker compose -f hadoop/docker-compose.yml ps | detect columns --guess
```

#	NAME	IMAGE	COMMAND	SERVICE	CREATED	STATUS	PORTS
0	namenode	registry.docker.ir/bde2020/hadoop-datanode:2.0.0-hadoop3.2.1-java8	"/entrypoint.sh sh -c"	namenode	3 minutes ago	Up 3 minutes (healthy)	0.0.0.0:9864->9864/tcp, [::]:9864->9864/tcp
1	namenode	registry.docker.ir/bde2020/hadoop-namenode:2.0.0-hadoop3.2.1-java8	"/entrypoint.sh sh -c"	namenode	3 minutes ago	Up 3 minutes (healthy)	0.0.0.0:9870->9870/tcp, [::]:9870->9870/tcp

شکل ۱۶: وضعیت کانترینرهای Hadoop



شکل ۱۷: رابط کاربری NameNode



شکل ۱۸: اجرای Job های Hadoop Streaming

`hdfs dfs -ls /input`

`hdfs dfs -ls /output`

```

root@a57ea740ff0c:/# hdfs dfs -ls /input
Found 2 items
-rw-r--r--  3 root supergroup  28882062 2026-07-10 07:50 /input/nginx_access.log
drwxr-xr-x  - root supergroup      0 2026-07-10 07:50 /input/service_logs
root@a57ea740ff0c:/#
root@a57ea740ff0c:/# hdfs dfs -ls /output
Found 5 items
drwxr-xr-x  - root supergroup      0 2026-07-10 07:50 /output/job1
drwxr-xr-x  - root supergroup      0 2026-07-10 07:51 /output/job2
drwxr-xr-x  - root supergroup      0 2026-07-10 07:51 /output/job3
drwxr-xr-x  - root supergroup      0 2026-07-10 07:51 /output/job4
drwxr-xr-x  - root supergroup      0 2026-07-10 07:51 /output/job5
root@a57ea740ff0c:/#

```

شکل ۱۹: محتویات HDFS قبل و بعد از پردازش

خروجی‌های میانی

پس از پردازش، فایل‌های زیر به‌عنوان خروجی‌های میانی تولید شدند:

توضیح	فایل	Job
لاگ دروازه‌ی تمیزشده، با request_time_ms	outputs/job1/cleanednginx_logs.csv	1
لاگ سرویس‌ها با entity_type/entity_value	outputs/job1/cleaned_service_logs.csv	1
رکوردهای نامعتبر	outputs/job1/invalid_logs.csv	1
تعداد، خطای error rate، 4xx/5xx و میانگین زمان پاسخ هر سرویس	outputs/job2/service_stats.csv	2
همان آمار به تفکیک endpoint	outputs/job2/endpoint_stats.csv	2
همان آمار به تفکیک سناریو	outputs/job2/scenario_stats.csv	2
تعداد درخواست هر کشور برای هر موجودیت	outputs/job3/country*requests.csv	3
محبوب‌ترین موجودیت هر کشور	outputs/job4/popular*by_country.csv	4

```
~/Desktop/cloud/CC_Project_Phase1_40131053_40131029/outputs> tree
.
├── final
│   └── summary.json
├── job1
│   ├── cleaned_nginx_logs.csv
│   ├── cleaned_service_logs.csv
│   └── invalid_logs.csv
├── job2
│   ├── endpoint_stats.csv
│   ├── scenario_stats.csv
│   └── service_stats.csv
├── job3
│   ├── country_city_requests.csv
│   ├── country_matchday_requests.csv
│   ├── country_stadium_requests.csv
│   └── country_team_requests.csv
└── job4
    ├── popular_city_by_country.csv
    ├── popular_matchday_by_country.csv
    ├── popular_stadium_by_country.csv
    └── popular_team_by_country.csv

5 directories, 15 files
```

شکل ۲۰: فایل‌های خروجی در پوشه‌ی outputs

نمونه‌ی outputs/job2/service_stats.csv:

```
~/Desktop/cloud/CC_Project_Phase1_40131053_40131029> cat outputs/job2/service_stats.csv | from csv
```

#	service	total_requests	success_count	error_4xx	error_5xx	error_rate	avg_response_time_ms
0	match-service	24564	21965	1592	1007	0.11	69.76
1	stadium-service	26378	21276	4261	841	0.19	165.28
2	team-service	36008	32563	1913	1532	0.10	47.02

شکل ۲۱: outputs/job2/service_stats.csv

نمونه‌ی outputs/job4/popular_team_by_country.csv:

```
~/Desktop/cloud/CC_Project_Phase1_40131053_40131029/outputs> cat job4/popular_team_by_country.csv | from csv
```

#	country	popular_team	total_requests
0	Argentina	Argentina	2488
1	Brazil	Brazil	2235
2	Canada	Canada	542
3	France	France	1572
4	Germany	Germany	2485
5	Iran	Argentina	3993
6	Japan	Japan	1226
7	Mexico	Mexico	1067
8	Morocco	Morocco	335
9	Spain	Spain	853
10	USA	USA	2180

شکل ۲۲: outputs/job4/popular_team_by_country.csv

خروجی نهایی — outputs/final/summary.json

```
~/Desktop/cloud/CC_Project_Phase1_40131053_40131029/outputs> cat final/summary.json | from json
```

service_stats	<table> <tr> <td>match-service</td><td> <table> <tr><td>error_rate</td><td>0.11</td></tr> <tr><td>error_5xx</td><td>1007</td></tr> <tr><td>avg_response_time_ms</td><td>69.76</td></tr> <tr><td>total_requests</td><td>24564</td></tr> <tr><td>success_count</td><td>21965</td></tr> <tr><td>error_4xx</td><td>1592</td></tr> </table> </td></tr> <tr> <td>stadium-service</td><td> <table> <tr><td>error_rate</td><td>0.19</td></tr> <tr><td>error_5xx</td><td>841</td></tr> <tr><td>avg_response_time_ms</td><td>165.28</td></tr> <tr><td>total_requests</td><td>26378</td></tr> <tr><td>success_count</td><td>21276</td></tr> <tr><td>error_4xx</td><td>4261</td></tr> </table> </td></tr> <tr> <td>team-service</td><td> <table> <tr><td>error_rate</td><td>0.10</td></tr> <tr><td>error_5xx</td><td>1532</td></tr> <tr><td>avg_response_time_ms</td><td>47.02</td></tr> <tr><td>total_requests</td><td>36008</td></tr> <tr><td>success_count</td><td>32563</td></tr> <tr><td>error_4xx</td><td>1913</td></tr> </table> </td></tr> </table>	match-service	<table> <tr><td>error_rate</td><td>0.11</td></tr> <tr><td>error_5xx</td><td>1007</td></tr> <tr><td>avg_response_time_ms</td><td>69.76</td></tr> <tr><td>total_requests</td><td>24564</td></tr> <tr><td>success_count</td><td>21965</td></tr> <tr><td>error_4xx</td><td>1592</td></tr> </table>	error_rate	0.11	error_5xx	1007	avg_response_time_ms	69.76	total_requests	24564	success_count	21965	error_4xx	1592	stadium-service	<table> <tr><td>error_rate</td><td>0.19</td></tr> <tr><td>error_5xx</td><td>841</td></tr> <tr><td>avg_response_time_ms</td><td>165.28</td></tr> <tr><td>total_requests</td><td>26378</td></tr> <tr><td>success_count</td><td>21276</td></tr> <tr><td>error_4xx</td><td>4261</td></tr> </table>	error_rate	0.19	error_5xx	841	avg_response_time_ms	165.28	total_requests	26378	success_count	21276	error_4xx	4261	team-service	<table> <tr><td>error_rate</td><td>0.10</td></tr> <tr><td>error_5xx</td><td>1532</td></tr> <tr><td>avg_response_time_ms</td><td>47.02</td></tr> <tr><td>total_requests</td><td>36008</td></tr> <tr><td>success_count</td><td>32563</td></tr> <tr><td>error_4xx</td><td>1913</td></tr> </table>	error_rate	0.10	error_5xx	1532	avg_response_time_ms	47.02	total_requests	36008	success_count	32563	error_4xx	1913
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success_count	32563																																										
error_4xx	1913																																										
total_4xx	7766																																										
most_popular_team_overall	Argentina																																										
predicted_final_winner	Argentina																																										
predicted_final	France vs Argentina																																										
most_requested_match_day_overall	2026-06-25																																										
popular_team_by_country	<table> <tr><td>Iran</td><td>Argentina</td></tr> <tr><td>Argentina</td><td>Argentina</td></tr> <tr><td>Mexico</td><td>Mexico</td></tr> <tr><td>Brazil</td><td>Brazil</td></tr> <tr><td>USA</td><td>USA</td></tr> <tr><td>Germany</td><td>Germany</td></tr> <tr><td>Spain</td><td>Spain</td></tr> <tr><td>Canada</td><td>Canada</td></tr> <tr><td>France</td><td>France</td></tr> <tr><td>Japan</td><td>Japan</td></tr> <tr><td>Morocco</td><td>Morocco</td></tr> </table>	Iran	Argentina	Argentina	Argentina	Mexico	Mexico	Brazil	Brazil	USA	USA	Germany	Germany	Spain	Spain	Canada	Canada	France	France	Japan	Japan	Morocco	Morocco																				
Iran	Argentina																																										
Argentina	Argentina																																										
Mexico	Mexico																																										
Brazil	Brazil																																										
USA	USA																																										
Germany	Germany																																										
Spain	Spain																																										
Canada	Canada																																										
France	France																																										
Japan	Japan																																										
Morocco	Morocco																																										

شکل ۲۳: محتوای فایل summary.json

تفسیر:

- **most_requested_service**: پرمراجعه‌ترین سرویس (از لاگ Nginx).
- **highest_error_rate_service**: سرویسی که بیشترین نسبت خطا دارد؛ چون ترافیک `invalid_input` بیشتر به سمت `stadium-service` هدایت شده، این سرویس بالاترین `error rate` را دارد.
- **slowest_endpoint: endpoint**: با بیشترین میانگین زمان پاسخ.
- **most_popular_team_overall / ..._match_day... / ..._stadium...** — از جمع خروجی Job 3 روی همه‌ی کشورها.
- **popular_team_by_country**: خروجی Job 4.
- **predicted_***: از داده‌ی پیش‌بینی مسابقات (`data/predictions/`).

بخش امتیازی — Spark Structured Streaming

برای پردازش زمان واقعی از Spark Streaming استفاده شد:

```
docker compose -f spark/docker-compose.yml run --rm spark /opt/spark/bin/spark-submit --master 'local[*]' /project/spark/streaming_app.py --input /project/data/stream/nginx --service-input /project/data/stream/service_logs --checkpoint /project/checkpoints/spark
```

```
~/Desktop/cloud/CC_Project_Phase1_40131053_40131029:
~/Desktop/cloud/CC_Project_Phase1_40131053_40131029:
~/Desktop/cloud/CC_Project_Phase1_40131053_40131029: docker compose -f spark/docker-compose.yml run --rm spark /opt/spark/bin/spark-submit --master 'local[*]' /project/spark/streaming_app.py --input /project/data/stream/nginx --service-input /project/data/stream/service_logs --checkpoint /project/checkpoints/spark
Container spark-spark-run-4d37cca2d5d6 Created
26/07/10 13:24:05 INFO SparkContext: Running Spark version 3.5.1
26/07/10 13:24:05 INFO SparkContext: OS info Linux, 6.8.0-110-generic, amd64
26/07/10 13:24:05 INFO SparkContext: Java version 11.0.22
26/07/10 13:24:05 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
26/07/10 13:24:05 INFO ResourceUtils: =====
26/07/10 13:24:05 INFO ResourceUtils: No custom resources configured for spark.driver.
26/07/10 13:24:05 INFO ResourceUtils: =====
26/07/10 13:24:05 INFO SparkContext: Submitted application: worldcup-live-log-analytics
26/07/10 13:24:05 INFO ResourceProfile: Default ResourceProfile created, executor resources: Map(cores -> name: cores, amount: 1, script: , vendor: , memory -> name: memory, amount: 1024, script: , vendor: , off
ap -> name: offheap, amount: 0, script: , vendor: ), task resources: Map(cpus -> name: cpus, amount: 1.0)
26/07/10 13:24:05 INFO ResourceProfile: Limiting resource is cpu
26/07/10 13:24:05 INFO ResourceProfileManager: Added ResourceProfile id: 0
26/07/10 13:24:05 INFO SecurityManager: Changing view acls to: root
26/07/10 13:24:05 INFO SecurityManager: Changing modify acls to: root
26/07/10 13:24:05 INFO SecurityManager: Changing view acls groups to:
26/07/10 13:24:05 INFO SecurityManager: Changing modify acls groups to:
26/07/10 13:24:05 INFO SecurityManager: SecurityManager: authentication disabled; ui acls disabled; users with view permissions: root; groups with view permissions: EMPTY; users with modify permissions: root; gr
ps with modify permissions: EMPTY
26/07/10 13:24:05 INFO Utils: Successfully started service 'sparkDriver' on port 33051.
26/07/10 13:24:05 INFO SparkEnv: Registering MapOutputTracker
26/07/10 13:24:05 INFO SparkEnv: Registering BlockManagerMaster
26/07/10 13:24:05 INFO BlockManagerMasterEndpoint: Using org.apache.spark.storage.DefaultTopologyMapper for getting topology information
26/07/10 13:24:05 INFO BlockManagerMasterEndpoint: BlockManagerMasterEndpoint up
26/07/10 13:24:05 INFO SparkEnv: Registering BlockManagerMasterHeartbeat
26/07/10 13:24:05 INFO DiskBlockManager: Created local directory at /tmp/blockmgr-0eadf09e-2138-48db-ae1a-16e02dde8ce0
26/07/10 13:24:05 INFO MemoryStore: MemoryStore started with capacity 434.4 MiB
26/07/10 13:24:05 INFO SparkEnv: Registering OutputCommitCoordinator
26/07/10 13:24:05 INFO JettyUtil: Start Jetty 9.4.0.v20180605 for SparkUI
26/07/10 13:24:05 INFO Utils: Successfully started service 'SparkUI' on port 4040.
26/07/10 13:24:05 INFO Executor: Starting executor ID driver on host da82b41569de
26/07/10 13:24:05 INFO Executor: OS info Linux, 6.8.0-110-generic, amd64
26/07/10 13:24:05 INFO Executor: Java version 11.0.22
26/07/10 13:24:05 INFO Executor: Starting executor with user classpath (userClassPathFirst = false): ''
26/07/10 13:24:05 INFO Executor: Created or updated repl class loader org.apache.spark.util.MutableURLClassLoader@4be31dca for default.
26/07/10 13:24:05 INFO Utils: Successfully started service 'org.apache.spark.network.netty.NettyBlockTransferService' on port 40307.
26/07/10 13:24:05 INFO NettyBlockTransferService: Server created on da82b41569de:40307
26/07/10 13:24:05 INFO BlockManager: Using org.apache.spark.storage.RandomBlockReplicationPolicy for block replication policy
26/07/10 13:24:05 INFO BlockManagerMaster: Registering BlockManager BlockManagerId(driver, da82b41569de, 40307, None)
26/07/10 13:24:05 INFO BlockManagerMasterEndpoint: Registering block manager da82b41569de:40307 with 424.4 MiB RAM, BlockManagerId(driver, da82b41569de, 40307, None)
26/07/10 13:24:05 INFO BlockManagerMaster: Registered BlockManager BlockManagerId(driver, da82b41569de, 40307, None)
26/07/10 13:24:05 INFO BlockManager: Initialized BlockManager: BlockManagerId(driver, da82b41569de, 40307, None)
26/07/10 13:24:07 WARN ResolveWriteToStream: spark.sql.adaptive.enabled is not supported in streaming DataFrames/Datasets and will be disabled.
26/07/10 13:24:07 WARN ResolveWriteToStream: spark.sql.adaptive.enabled is not supported in streaming DataFrames/Datasets and will be disabled.
26/07/10 13:24:07 WARN ResolveWriteToStream: spark.sql.adaptive.enabled is not supported in streaming DataFrames/Datasets and will be disabled.
26/07/10 13:24:07 WARN ResolveWriteToStream: spark.sql.adaptive.enabled is not supported in streaming DataFrames/Datasets and will be disabled.
Streaming queries started: windowed service stats, live error rate, busiest endpoint, popular_team_by_country_live
Drop new *.jsonl files into /project/data/stream/nginx to see the numbers update.
```

شکل ۲۴: ساخت docker compose

Busiest service / endpoint (micro-batch 16)					
window_start	window_end	service	requests	avg_response_time_ms	error_rate_pct
2026-07-09 21:30:10	2026-07-09 21:30:20	team-service	335	69.4	8.96
2026-07-09 21:30:10	2026-07-09 21:30:20	match-service	223	71.1	9.42
2026-07-09 21:30:00	2026-07-09 21:30:10	team-service	1432	42.6	9.5
2026-07-09 21:30:00	2026-07-09 21:30:10	match-service	983	63.9	11.09
2026-07-09 21:30:10	2026-07-09 21:30:20	stadium-service	236	166.9	19.07
2026-07-09 21:30:00	2026-07-09 21:30:10	stadium-service	1036	179.9	19.02
service	endpoint	requests			
team-service	/api/teams	2815			
stadium-service	/api/stadiums	2032			
match-service	/api/matches	1952			
nginx	/health	1			

شکل ۲۵: اجرای برنامه‌ی Spark Streaming